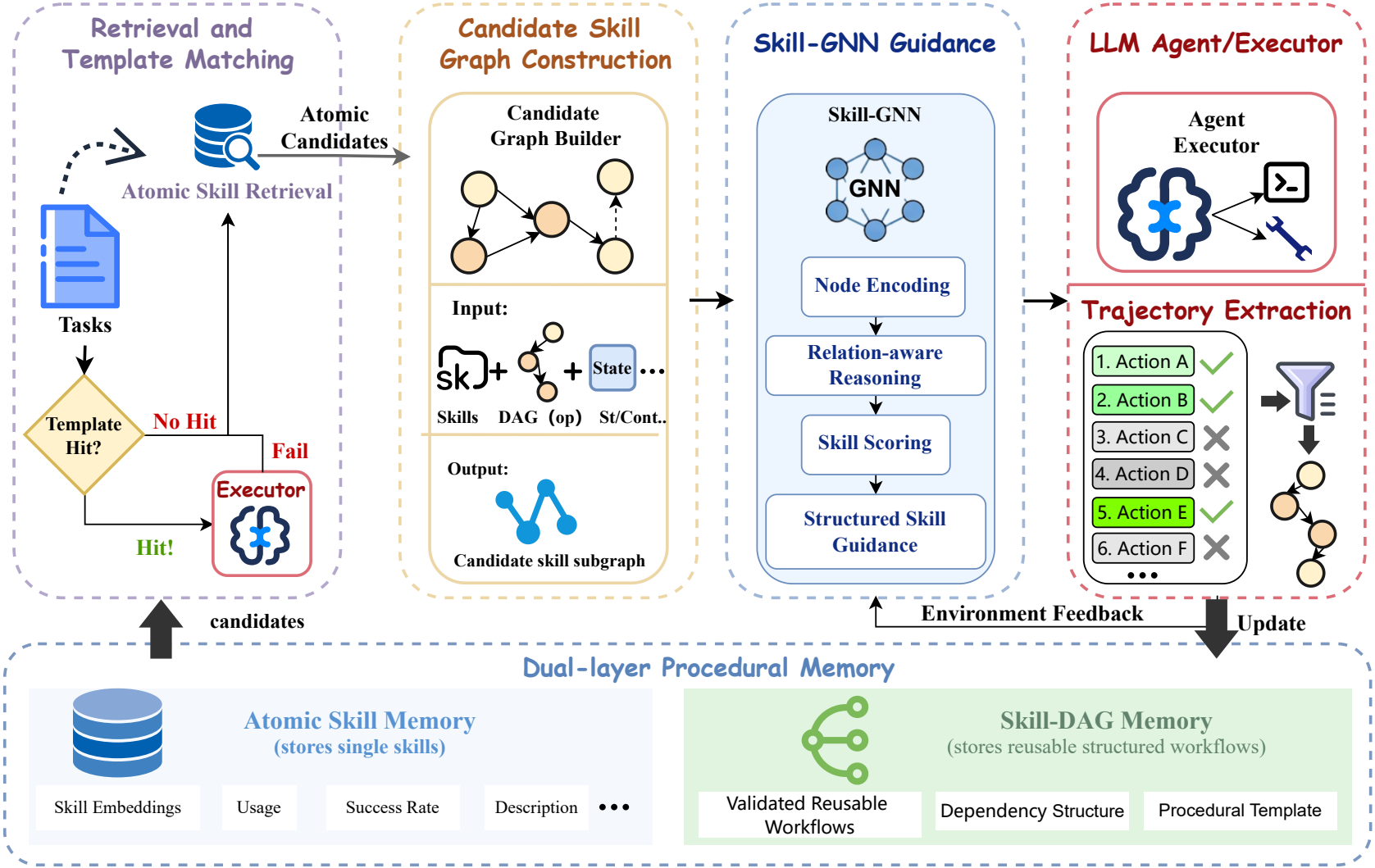

Retrieval and Template Matching
Atomic Candidates
Atomic Skill Retrieval
Tasks
Template Hit?
No Hit
Fail
Executor
Hit!
candidates
Candidate Skill Graph Construction
Candidate Graph Builder
Input:
Skills
DAG (op)
St/Cont..
State
Output:
Candidate skill subgraph
Skill-GNN Guidance
Skill-GNN
GNN
Node Encoding
Relation-aware Reasoning
Skill Scoring
Structured Skill Guidance
LLM Agent/Executor
Agent Executor
Trajectory Extraction
1. Action A
2. Action B
3. Action C
4. Action D
5. Action E
6. Action F
Environment Feedback
Update
Dual-layer Procedural Memory
Atomic Skill Memory
(stores single skills)
Skill Embeddings
Usage
Success Rate
Description
Skill-DAG Memory
(stores reusable structured workflows)
Validated Reusable Workflows
Dependency Structure
Procedural Template